

EXP.NO 2

Demonstrate the statistics for a sample data like mean, standard deviation, normal/uniform distribution, variance and correlation.

AIM:

To demonstrate the statistics for a sample data like mean, standard deviation, normal/uniform distribution, variance and correlation.

ALGORITHM:

- Import required libraries (NumPy, Pandas, Matplotlib)
- Generate sample dataset
- Compute:
 - Mean
 - Standard Deviation
 - Variance
- Generate:
 - Normal distribution data
 - Uniform distribution data
- Calculate correlation between variables
- Display results and plots

SOURCE CODE:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
A = np.array([60, 70, 80, 90, 100])
B = np.array([1, 2, 3, 4, 5])
U = np.array([1, 2, 3, 4, 5])
mean_A = np.mean(A)
variance_A = np.var(A)
std_dev_A = np.std(A)
correlation_AB = np.corrcoef(A, B)[0, 1]
print("Dataset A:", A)
print("Mean:", mean_A)
print("Variance:", variance_A)
print("Standard Deviation:", std_dev_A)
print("Correlation between A and B:", correlation_AB)
x = np.linspace(mean_A - 3*std_dev_A, mean_A + 3*std_dev_A, 100)
normal_dist = stats.norm.pdf(x, mean_A, std_dev_A)
plt.figure(figsize=(12, 4))
plt.subplot(1, 3, 1)
plt.hist(A, bins=5, density=True, alpha=0.6, color='skyblue')
plt.plot(x, normal_dist, 'r', linewidth=2)
plt.title("Normal Distribution (Dataset A)")
```

```

plt.xlabel("Values")
plt.ylabel("Density")
plt.subplot(1, 3, 2)
plt.hist(U, bins=5, density=True, alpha=0.6, color='lightgreen')
plt.title("Uniform Distribution")
plt.xlabel("Values")
plt.ylabel("Density")
plt.subplot(1, 3, 3)
plt.scatter(B, A, color='orange')
plt.title("Correlation between A and B")
plt.xlabel("Dataset B")
plt.ylabel("Dataset A")
plt.tight_layout()
plt.show()

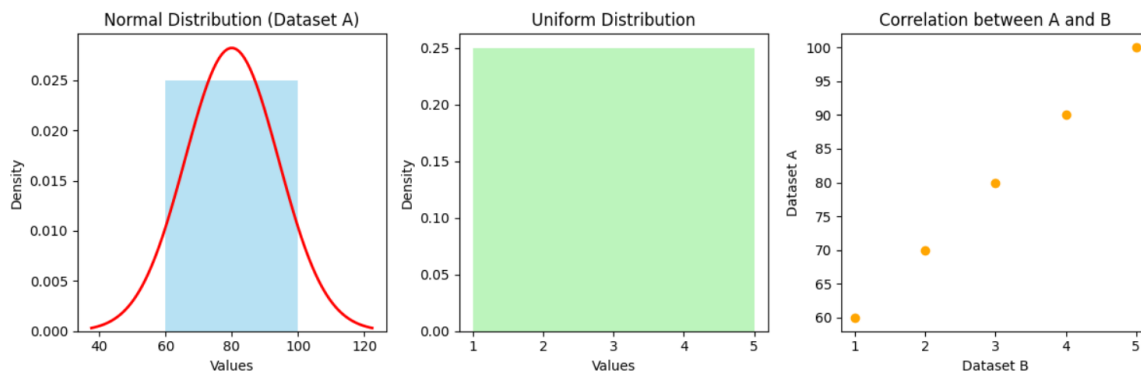
```

OUTPUT:

```

Dataset A: [ 60  70  80  90 100]
Mean: 80.0
Variance: 200.0
Standard Deviation: 14.142135623730951
Correlation between A and B: 1.0

```



RESULT:

Thus, the statistics for a sample data like mean, standard deviation, normal/uniform distribution, variance and correlation successfully executed.

